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3/3/2021

$$r = \eta \times R_f$$

$$r = 0.41142 \text{ cm}$$

$$1. a. LHR = 300 \text{ W/cm}$$

$$r_f = 0.55 \text{ cm}$$

$$k_R = 0.025 \text{ W/cm-K}$$

$$E = 200 \text{ GPa}$$

$$\nu = 0.35$$

$$\alpha = 10 \times 10^{-6} \text{ 1/K}$$

$$\eta = \frac{r}{r_f}$$

$$r = r_f$$

$$\eta = 1$$

$$\sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1-\nu)}$$

$$T_0 - T_s = \frac{LHR}{4\pi k_R}$$

$$T_0 - T_s = 954.93 \text{ K}$$

$$\sigma^* = 734.56 \text{ MPa}$$

$$\sigma_z(\eta) = -2\sigma^*(1-2\eta^2)$$

$$\sigma_z = 1.46912 \text{ GPa}$$

→ should be σ_θ , but $\sigma_\theta = \sigma_z$ here

$$b. \sigma_{fracture} = 175 \text{ MPa}$$

$$\sigma_z(\eta) = -2\sigma^*(1-2\eta^2)$$

right process, wrong σ

$$-\frac{\sigma_z(\eta)}{2\sigma^*} = 1 - 2\eta^2$$

$$\frac{\sigma_z(\eta)}{4\sigma^*} + \frac{1}{2} = \eta^2$$

$$\eta = \sqrt{\frac{\sigma_{fracture}}{4\sigma^*} + \frac{1}{2}}$$

$$\eta = 0.75$$

2. a. $P = 15 \text{ MPa}$
 $R_c = 0.55 \text{ cm}$
 $\delta_c = 0.05 \text{ cm}$
 $\frac{R_c}{\delta_c} = 11$

$11 < 20$

thin-walled

$R_i = 0.525 \text{ cm}$

$R_o = 0.575 \text{ cm}$ $R = R_o$

$\sigma_\theta = \frac{PR}{\delta}$

$\sigma_\theta = 165 \text{ MPa}$

$\sigma_z = \frac{PR}{2\delta}$

$\sigma_z = 82.5 \text{ MPa}$

$\sigma_r = -\frac{P}{2}$

$\sigma_r = -7.5 \text{ MPa}$

b. $r = R_i$

$\sigma_r(r) = -P \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1}$

$\sigma_r(r) = -15 \text{ MPa}$

$\sigma_\theta(r) = +P \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1}$

$\sigma_\theta(r) = +165.34091 \text{ MPa}$

$\sigma_z(r) = \frac{P}{(R_o/R_i)^2 - 1}$

$\sigma_z(r) = 75.17045 \text{ MPa}$

3. $R_F = 0.48 \text{ cm}$

$t_g = 0.008 \text{ cm}$

$T_{co} = 550 \text{ K}$

$t_c = 0.08 \text{ cm}$

$k_F = 0.03 \text{ W/cm-K}$

$k_g = 0.003 \text{ W/cm-K}$

$k_c = 0.15 \text{ W/cm-K}$

$LHR = 250 \text{ W/cm}$

$\alpha_c = 10 \times 10^{-6} \text{ 1/K}$

$\alpha_F = 14 \times 10^{-6} \text{ 1/K}$

$T_{ref} = 300 \text{ K}$

$T_{ci} = T_{co} + \frac{LHR t_c}{2\pi R_F k_c}$

$T_{ci} = 594.21 \text{ K}$

$T_s = T_{ci} + \frac{LHR t_g}{2\pi R_F k_g}$

$T_s = 815.26 \text{ K}$

$T_o = T_s + \frac{LHR}{4\pi k_F}$

$T_o = 1,478.40 \text{ K}$

$R_c = R_F + t_g + \frac{t_c}{2}$

$R_c = 0.528 \text{ cm}$

$T_c = \frac{T_o + T_{ci}}{2}$

$T_c = 572.10 \text{ K}$

$T_F = \frac{T_s + T_o}{2}$

$T_F = 1,146.83 \text{ K}$

$\Delta S_{gap} = R_c \alpha_c (T_c - T_{ref}) + R_F \alpha_F (T_o - T_{ref})$

$\Delta S_{gap} = 7.12742 \times 10^{-3} \text{ cm}$

- everything looked right, but wrong answer... show values into last eqn.

$$4. \Delta T = 50 \text{ K}$$

$$\alpha_c = 15 \times 10^{-6}$$

$$E = 100 \text{ GPa}$$

$$\nu = 0.34$$

$$t_c = 0.06 \text{ cm}$$

$$R_i = 0.55 \text{ cm}$$

$$r = R_i$$

$$\sigma_r(r) = \frac{3}{2} \Delta T \frac{\alpha E}{1-\nu} \left(\frac{r_i}{r} - 1 \right) \left(1 - \frac{R_i}{r} \left(\frac{r_i}{r} - 1 \right) \right)$$

$$\boxed{\sigma_r(r) = 0 \text{ MPa}} \quad \checkmark$$

$$\sigma_\theta(r) = \frac{3}{2} \Delta T \frac{\alpha E}{1-\nu} \left(1 - 2 \frac{R_i}{r} \left(\frac{r_i}{r} - 1 \right) \right)$$

$$\boxed{\sigma_\theta(r) = 0 \text{ MPa}} \quad \times \text{ - this doesn't go to zero!}$$

$$5. \dot{F} = 6 \times 10^{13} \text{ fissions/cm}^3\text{-s}$$

$$A_{po} = 0.016$$

$$B_0 = 5 \text{ MW/MkgU}$$

$$\rho = 10.97 \text{ g/cc}$$

$$t = 10 \text{ d}$$

$$C_0 = 1 \quad M_U = 270 \text{ g/mol}$$

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

$$N_U = \rho \frac{N_A}{M_U} \quad \checkmark$$

$$N_U = 2.45 \times 10^{22} \text{ atoms} \quad \checkmark \rightarrow \text{want to see math}$$

$$B = \frac{\dot{F} t}{N_U}$$

$$B = 0.0021 \text{ FIMA} \quad \checkmark$$

$$B_0 = 0.0053 \text{ FIMA} \quad \checkmark$$

$$\epsilon_D = \Delta \rho_0 \left(\frac{B_0 \alpha \rho_0}{G_0 H_0} - 1 \right)$$

$$\boxed{\epsilon_D = -0.012651} \quad \checkmark$$

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6. Elasticity is a material's ability to undergo stress without having permanent deformation when that stress is no longer applied.
- Plasticity is a material's ability to undergo stress after permanent deformation and not fracture.
- Kind of, mostly*
7. Strain hardening is when a material hardens due to stress being applied to it. *but what does this mean?* It occurs when plastic deformation occurs and permanent strain occurs. It is caused by dislocations being formed and hardening the material. *- how?*
8. 1. Numerically model the fuel's temperature -
 2. Numerically model the stresses on the cladding -
 3. Model the changes in gap pressure, temperature, and size
9. Grains form during the sintering process when two pieces of powder combine together. The unaligned crystalline structure of the pieces form grains.
10. Microstructure-based mechanistic fuel modeling is modeling the fuel at the microscopic level rather than due to burnup. This provides more detailed models, particularly for fuels with limited experimental data. *- wanted a bit more*
11. Microstructure is what is seen at 25X magnification. Annealing is a technique where the material is heated and then cooled slowly to let the material "get rid of" defects such as dislocations, voids, and interstitials.
12. It occurs more on the grain boundaries. There is more space in the grain boundaries.
13. Temperature is a driving force for grain growth. Defects like voids can impede grain growth. *- temperature allows it, but T isn't the driving force*

14. Fuel densifies due to burnup. This process involves thermal expansion^x of the fuel which causes the pores in the fuel to collapse. This creates a denser fuel. ^{2/6} -not really

